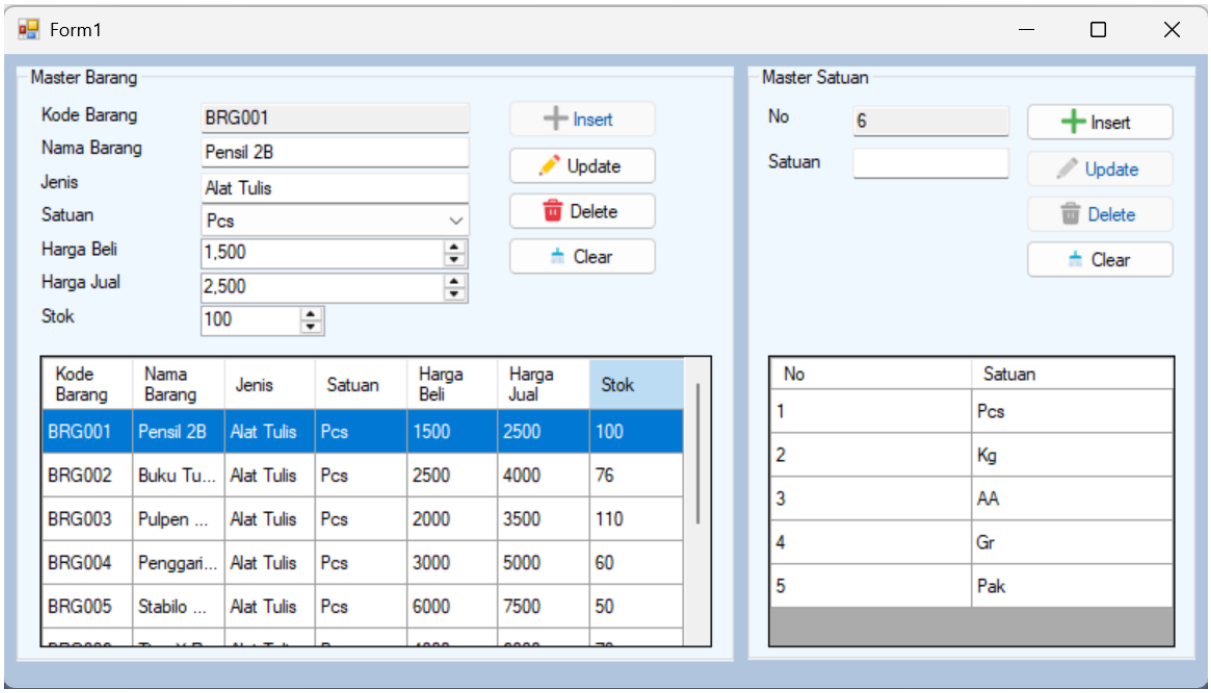
Desktop Programming – Week 10

Individual Practice

1. Create Form

Create a form like the one below:



Input components and buttons

Master Barang

Name in Form	Toolbox	Properties Name
Kode barang automatic (read-only)	TextBox	txtKodeBarang
Nama barang	TextBox	txtNamaBarang
Jenis barang	TextBox	txtJenis
Satuan	ComboBox	cmbSatuan
Harga beli	NumericUpDown	numHargaBeli
Harga jual	NumericUpDown	numHargaJual
Stok	NumericUpDown	numStock
Insert	Button	btnSimpanBarang
Update	Button	btnUbahBarang
Delete	Button	btnHapusBarang
Clear	Button	btnBersihBarang
List of all items	DataGridView	dgvBarang

Master Satuan

Name in Form	Toolbox	Properties Name
Nomor satuan automatic (read-only)	TextBox	txtNoSatuan
Nama satuan	TextBox	txtSatuan
Insert	Button	btnSimpanSatuan
Update	Button	btnUbahSatuan
Delete	Button	btnHapusSatuan
Clear	Button	btnBersihSatuan
Displaying unit master	DataGridView	dgvSatuan

Set button properties

Change the 'Text' property for the eight buttons to Insert, Update, 'Delete', and 'Clear', corresponding to the 'Barang' or 'Satuan' sections. Button properties:
- Image : Resouce context → choose from local resource (.PNG and 16 x16 px)
- ImageAlign : MiddleCenter
- TextAlign : MiddleRight
- TextImageRelation : ImageBeforeText

2. **Database**
   Reuse the existing tblsatuan and tblbarang tables. Install MySql.Data into your project.
3. **Code**
   Adjust the code for the database connection according to your database configuration.

```
Option Strict On
Option Explicit On

Imports System
Imports System.Data
Imports System.Drawing
Imports System.Windows.Forms
Imports MySql.Data.MySqlClient

Public Class Form1
    Private Const CONNECTION_STRING As String =
        "Server=localhost;Port=3306;Database=dbtrial;Uid=root;Pwd=root;CharSet=utf8mb4;"

    Private Const TABLE_SATUAN As String = "tblSatuan"
    Private _kodeBarangTerpilih As String = String.Empty
    Private _noSatuanTerpilih As Integer = -1
    Private _satuanLama As String = String.Empty

    Private Function BukaKoneksi() As MySqlConnection
        Dim koneksi As New MySqlConnection(CONNECTION_STRING)
        koneksi.Open()
        Return koneksi
    End Function

    Private Sub Form_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        Try
            KonfigurasiTampilan()
            MuatSatuan()
```

```
        MuatBarang()
        MuatGridSatuan()
        BersihkanBarang()
        BersihkanSatuan()
    Catch ex As Exception
        TampilkanError("Gagal menyiapkan form", ex)
    End Try
End Sub

Private Sub KonfigurasiTampilan()
    txtKodeBarang.ReadOnly = True
    txtNoSatuan.ReadOnly = True
    txtNamaBarang.MaxLength = 50
    txtJenis.MaxLength = 50
    txtSatuan.MaxLength = 50
    cmbSatuan.DropDownStyle = ComboBoxStyle.DropDownList

    For Each angka As NumericUpDown In New NumericUpDown() {numHargaBeli, numHargaJual, numStock}
        angka.Minimum = 0D
        angka.Maximum = CDec(Integer.MaxValue)
        angka.DecimalPlaces = 0
        angka.ThousandsSeparator = True
    Next

    For Each grid As DataGridView In New DataGridView() {dgvBarang, dgvSatuan}
        grid.ReadOnly = True
        grid.AllowUserToAddRows = False
        grid.AllowUserToDeleteRows = False
        grid.MultiSelect = False
        grid.SelectionMode = DataGridViewSelectionMode.FullRowSelect
        grid.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill
        grid.RowHeadersVisible = False
    Next
End Sub

' ------------------------- DATA BARANG -------------------------

Private Function AmbilKodeBarangBerikutnya(koneksi As MySqlConnection) As String
    Const sql As String =
        "SELECT COALESCE(MAX(CAST(SUBSTRING(Kode_Barang, 4) AS UNSIGNED)), 0) " &
        "FROM tblbarang WHERE Kode_Barang REGEXP '^BRG[0-9]+$'"
    Using cmd As New MySqlCommand(sql, koneksi)
        Dim nomor As Long = Convert.ToInt64(cmd.ExecuteScalar()) + 1L
        Return "BRG" & nomor.ToString("D3")
    End Using
End Function

Private Function BuatKodeBarang() As String
    Using koneksi As MySqlConnection = BukaKoneksi()
        Return AmbilKodeBarangBerikutnya(koneksi)
    End Using
End Function

Private Sub MuatSatuan(Optional pilihan As String = "")
    Dim tabel As New DataTable()
    Using koneksi As MySqlConnection = BukaKoneksi()
        Dim sql As String = "SELECT `no`, satuan FROM `" & TABLE_SATUAN & "` ORDER BY satuan"
        Using adapter As New MySqlDataAdapter(sql, koneksi)
            adapter.Fill(tabel)
        End Using
    End Using

    cmbSatuan.DataSource = Nothing
    cmbSatuan.DisplayMember = "satuan"
    cmbSatuan.ValueMember = "satuan"
    cmbSatuan.DataSource = tabel
    cmbSatuan.SelectedIndex = -1

    If pilihan <> String.Empty Then
        cmbSatuan.SelectedValue = pilihan
    End If
End Sub

Private Sub MuatBarang()
```

```
    Dim tabel As New DataTable()
    Const sql As String =
        "SELECT Kode_Barang, Nama_Barang, Jenis, Satuan, Harga_Beli, Harga_Jual, Stock " &
        "FROM tblbarang ORDER BY Kode_Barang"

    Using koneksi As MySqlConnection = BukaKoneksi()
        Using adapter As New MySqlDataAdapter(sql, koneksi)
            adapter.Fill(tabel)
        End Using
    End Using

    dgvBarang.DataSource = tabel
    AturJudulKolom(dgvBarang, "Kode_Barang", "Kode Barang")
    AturJudulKolom(dgvBarang, "Nama_Barang", "Nama Barang")
    AturJudulKolom(dgvBarang, "Jenis", "Jenis")
    AturJudulKolom(dgvBarang, "Satuan", "Satuan")
    AturJudulKolom(dgvBarang, "Harga_Beli", "Harga Beli")
    AturJudulKolom(dgvBarang, "Harga_Jual", "Harga Jual")
    AturJudulKolom(dgvBarang, "Stock", "Stok")
    dgvBarang.ClearSelection()
End Sub

Private Sub AturJudulKolom(grid As DataGridView, kolom As String, judul As String)
    If grid.Columns.Contains(kolom) Then
        grid.Columns(kolom).HeaderText = judul
    End If
End Sub

Private Sub BersihkanBarang()
    _kodeBarangTerpilih = String.Empty
    txtKodeBarang.Text = BuatKodeBarang()
    txtNamaBarang.Clear()
    txtJenis.Clear()
    cmbSatuan.SelectedIndex = -1
    numHargaBeli.Value = 0D
    numHargaJual.Value = 0D
    numStock.Value = 0D
    btnSimpanBarang.Enabled = True
    btnUbahBarang.Enabled = False
    btnHapusBarang.Enabled = False
    dgvBarang.ClearSelection()
End Sub

Private Function ValidasiBarang() As Boolean
    If txtNamaBarang.Text.Trim() = "" Then
        MessageBox.Show("Nama barang wajib diisi.", "Validasi")
        txtNamaBarang.Focus()
        Return False
    End If
    If txtJenis.Text.Trim() = "" Then
        MessageBox.Show("Jenis barang wajib diisi.", "Validasi")
        txtJenis.Focus()
        Return False
    End If
    If cmbSatuan.SelectedIndex < 0 Then
        MessageBox.Show("Pilih satuan barang.", "Validasi")
        cmbSatuan.Focus()
        Return False
    End If
    Return True
End Function

Private Sub TambahkanParameterBarang(cmd As MySqlCommand)
    cmd.Parameters.Add("@nama", MySqlDbType.VarChar, 50).Value = txtNamaBarang.Text.Trim()
    cmd.Parameters.Add("@jenis", MySqlDbType.VarChar, 50).Value = txtJenis.Text.Trim()
    cmd.Parameters.Add("@satuan", MySqlDbType.VarChar, 50).Value = CStr(cmbSatuan.SelectedValue)
    cmd.Parameters.Add("@beli", MySqlDbType.Int32).Value = Decimal.ToInt32(numHargaBeli.Value)
    cmd.Parameters.Add("@jual", MySqlDbType.Int32).Value = Decimal.ToInt32(numHargaJual.Value)
    cmd.Parameters.Add("@stock", MySqlDbType.Int32).Value = Decimal.ToInt32(numStock.Value)
End Sub

Private Sub btnSimpanBarang_Click(sender As Object, e As EventArgs) Handles btnSimpanBarang.Click
    If _kodeBarangTerpilih <> String.Empty Then
        MessageBox.Show("Anda sedang mengedit barang. Klik Ubah untuk langsung menyimpan perubahan.", "Informasi")
```

```
        Return
    End If
    If Not ValidasiBarang() Then Return

    Const sql As String =
        "INSERT INTO tblbarang (Kode_Barang, Nama_Barang, Jenis, Satuan, Harga_Beli, Harga_Jual, Stock) " &
        "VALUES (@kode, @nama, @jenis, @satuan, @beli, @jual, @stock)"
    Try
        ' Coba ulang jika ada user lain yang lebih dulu menyimpan kode yang sama.
        Using koneksi As MySqlConnection = BukaKoneksi()
            Dim berhasil As Boolean = False
            For percobaan As Integer = 1 To 5
                Dim kodeBaru As String = AmbilKodeBarangBerikutnya(koneksi)
                Try
                    Using cmd As New MySqlCommand(sql, koneksi)
                        cmd.Parameters.Add("@kode", MySqlDbType.VarChar, 50).Value = kodeBaru
                        TambahkanParameterBarang(cmd)
                        cmd.ExecuteNonQuery()
                    End Using
                    txtKodeBarang.Text = kodeBaru
                    berhasil = True
                    Exit For
                Catch ex As MySqlException When ex.Number = 1062 AndAlso percobaan < 5
                    ' Duplicate primary key: ambil kode terbaru dan coba kembali.
                End Try
            Next
            If Not berhasil Then
                Throw New InvalidOperationException("Kode barang bentrok. Silakan coba Simpan kembali.")
            End If
        End Using

        MessageBox.Show("Barang berhasil disimpan.", "Informasi")
        MuatBarang()
        BersihkanBarang()
    Catch ex As Exception
        TampilkanError("Gagal menyimpan barang", ex)
    End Try
End Sub

Private Sub btnUbahBarang_Click(sender As Object, e As EventArgs) Handles btnUbahBarang.Click
    If _kodeBarangTerpilih = "" Then
        MessageBox.Show("Pilih barang di tabel terlebih dahulu.", "Informasi")
        Return
    End If
    If Not ValidasiBarang() Then Return

    Const sql As String =
        "UPDATE tblbarang SET Nama_Barang=@nama, Jenis=@jenis, Satuan=@satuan, " &
        "Harga_Beli=@beli, Harga_Jual=@jual, Stock=@stock WHERE Kode_Barang=@kode"
    Try
        Using koneksi As MySqlConnection = BukaKoneksi()
            Using cmd As New MySqlCommand(sql, koneksi)
                cmd.Parameters.Add("@kode", MySqlDbType.VarChar, 50).Value = _kodeBarangTerpilih
                TambahkanParameterBarang(cmd)
                If cmd.ExecuteNonQuery() = 0 Then
                    MessageBox.Show("Tidak ada data yang berubah atau barang tidak ditemukan.", "Informasi")
                Else
                    MessageBox.Show("Barang berhasil diubah.", "Informasi")
                End If
            End Using
        End Using
        MuatBarang()
        BersihkanBarang()
    Catch ex As Exception
        TampilkanError("Gagal mengubah barang", ex)
    End Try
End Sub

Private Sub btnHapusBarang_Click(sender As Object, e As EventArgs) Handles btnHapusBarang.Click
    If _kodeBarangTerpilih = "" Then
        MessageBox.Show("Pilih barang di tabel terlebih dahulu.", "Informasi")
        Return
    End If
    If MessageBox.Show("Hapus barang " & _kodeBarangTerpilih & "?", "Konfirmasi",
```

```
                MessageBoxButtons.YesNo, MessageBoxIcon.Question) <> DialogResult.Yes Then Return
    Try
        Using koneksi As MySqlConnection = BukaKoneksi()
            Using cmd As New MySqlCommand("DELETE FROM tblbarang WHERE Kode_Barang=@kode", koneksi)
                cmd.Parameters.Add("@kode", MySqlDbType.VarChar, 50).Value = _kodeBarangTerpilih
                cmd.ExecuteNonQuery()
            End Using
        End Using
        MuatBarang()
        BersihkanBarang()
        MessageBox.Show("Barang berhasil dihapus.", "Informasi")
    Catch ex As Exception
        TampilkanError("Gagal menghapus barang", ex)
    End Try
End Sub

Private Sub btnBersihBarang_Click(sender As Object, e As EventArgs) Handles btnBersihBarang.Click
    Try
        BersihkanBarang()
    Catch ex As Exception
        TampilkanError("Gagal mengosongkan isian", ex)
    End Try
End Sub

Private Sub dgvBarang_CellClick(sender As Object, e As DataGridViewCellEventArgs) Handles dgvBarang.CellClick
    If e.RowIndex < 0 Then Return
    Dim baris As DataGridViewRow = dgvBarang.Rows(e.RowIndex)
    If baris.IsNewRow Then Return
    Try
        _kodeBarangTerpilih = Convert.ToString(baris.Cells("Kode_Barang").Value)
        txtKodeBarang.Text = _kodeBarangTerpilih
        txtNamaBarang.Text = Convert.ToString(baris.Cells("Nama_Barang").Value)
        txtJenis.Text = Convert.ToString(baris.Cells("Jenis").Value)
        cmbSatuan.SelectedValue = Convert.ToString(baris.Cells("Satuan").Value)
        numHargaBeli.Value = Convert.ToDecimal(baris.Cells("Harga_Beli").Value)
        numHargaJual.Value = Convert.ToDecimal(baris.Cells("Harga_Jual").Value)
        numStock.Value = Convert.ToDecimal(baris.Cells("Stock").Value)
        ' Pilihan baris masuk mode edit: tombol Ubah langsung menyimpan UPDATE.
        btnSimpanBarang.Enabled = False
        btnUbahBarang.Enabled = True
        btnHapusBarang.Enabled = True
    Catch ex As Exception
        TampilkanError("Gagal membaca baris barang", ex)
    End Try
End Sub

' -------------------------- DATA SATUAN --------------------------

Private Function AmbilNomorSatuanBerikutnya(koneksi As MySqlConnection) As Integer
    Dim sql As String = "SELECT COALESCE(MAX(`no`),0)+1 FROM `" & TABLE_SATUAN & "`"
    Using cmd As New MySqlCommand(sql, koneksi)
        Return Convert.ToInt32(cmd.ExecuteScalar())
    End Using
End Function

Private Sub MuatGridSatuan()
    Dim tabel As New DataTable()
    Using koneksi As MySqlConnection = BukaKoneksi()
        Dim sql As String = "SELECT `no`, satuan FROM `" & TABLE_SATUAN & "` ORDER BY `no`"
        Using adapter As New MySqlDataAdapter(sql, koneksi)
            adapter.Fill(tabel)
        End Using
    End Using
    dgvSatuan.DataSource = tabel
    AturJudulKolom(dgvSatuan, "no", "No")
    AturJudulKolom(dgvSatuan, "satuan", "Satuan")
    dgvSatuan.ClearSelection()
End Sub

Private Sub BersihkanSatuan()
    _noSatuanTerpilih = -1
    _satuanLama = String.Empty
    txtSatuan.Clear()
    btnSimpanSatuan.Enabled = True
```

```
    btnUbahSatuan.Enabled = False
    btnHapusSatuan.Enabled = False
    Using koneksi As MySqlConnection = BukaKoneksi()
        txtNoSatuan.Text = AmbilNomorSatuanBerikutnya(koneksi).ToString()
    End Using
    dgvSatuan.ClearSelection()
End Sub

Private Function ValidasiSatuan() As Boolean
    If txtSatuan.Text.Trim() = "" Then
        MessageBox.Show("Nama satuan wajib diisi.", "Validasi")
        txtSatuan.Focus()
        Return False
    End If
    Return True
End Function

Private Function SatuanSudahAda(koneksi As MySqlConnection,
                                nama As String,
                                Optional kecualiNo As Integer = -1,
                                Optional transaksi As MySqlTransaction = Nothing) As Boolean
    Dim sql As String = "SELECT COUNT(*) FROM `" & TABLE_SATUAN & "` " &
                        "WHERE LOWER(satuan)=LOWER(@nama) AND `no`<>@kecuali"
    Using cmd As New MySqlCommand(sql, koneksi, transaksi)
        cmd.Parameters.Add("@nama", MySqlDbType.VarChar, 50).Value = nama
        cmd.Parameters.Add("@kecuali", MySqlDbType.Int32).Value = kecualiNo
        Return Convert.ToInt32(cmd.ExecuteScalar()) > 0
    End Using
End Function

Private Sub btnSimpanSatuan_Click(sender As Object, e As EventArgs) Handles btnSimpanSatuan.Click
    If _noSatuanTerpilih >= 0 Then
        MessageBox.Show("Anda sedang mengedit satuan. Klik Ubah untuk langsung menyimpan perubahan.", "Informasi")
        Return
    End If
    If Not ValidasiSatuan() Then Return
    Try
        Using koneksi As MySqlConnection = BukaKoneksi()
            If SatuanSudahAda(koneksi, txtSatuan.Text.Trim()) Then
                MessageBox.Show("Satuan tersebut sudah terdaftar.", "Validasi")
                Return
            End If
            Dim sql As String = "INSERT INTO `" & TABLE_SATUAN & "` (`no`, satuan) VALUES (@no, @nama)"
            Using cmd As New MySqlCommand(sql, koneksi)
                cmd.Parameters.Add("@no", MySqlDbType.Int32).Value = AmbilNomorSatuanBerikutnya(koneksi)
                cmd.Parameters.Add("@nama", MySqlDbType.VarChar, 50).Value = txtSatuan.Text.Trim()
                cmd.ExecuteNonQuery()
            End Using
        End Using
        Dim dipilih As String = txtSatuan.Text.Trim()
        MuatGridSatuan()
        MuatSatuan(dipilih)
        BersihkanSatuan()
        MessageBox.Show("Satuan berhasil disimpan.", "Informasi")
    Catch ex As Exception
        TampilkanError("Gagal menyimpan satuan", ex)
    End Try
End Sub

Private Sub btnUbahSatuan_Click(sender As Object, e As EventArgs) Handles btnUbahSatuan.Click
    If _noSatuanTerpilih < 0 Then
        MessageBox.Show("Pilih satuan di tabel terlebih dahulu.", "Informasi")
        Return
    End If
    If Not ValidasiSatuan() Then Return
    Dim namaBaru As String = txtSatuan.Text.Trim()
    Try
        Using koneksi As MySqlConnection = BukaKoneksi()
            Using transaksi As MySqlTransaction = koneksi.BeginTransaction()
                Try
                    If SatuanSudahAda(koneksi, namaBaru, _noSatuanTerpilih, transaksi) Then
                        MessageBox.Show("Nama satuan sudah digunakan.", "Validasi")
                        transaksi.Rollback()
                        Return
```

```
                End If

                Dim sql As String = "UPDATE `" & TABLE_SATUAN & "` SET satuan=@baru WHERE `no`=@no"
                Using cmd As New MySqlCommand(sql, koneksi, transaksi)
                    cmd.Parameters.Add("@baru", MySqlDbType.VarChar, 50).Value = namaBaru
                    cmd.Parameters.Add("@no", MySqlDbType.Int32).Value = _noSatuanTerpilih
                    cmd.ExecuteNonQuery()
                End Using

                ' Kolom tblbarang.Satuan berupa teks, jadi perbarui juga referensinya.
                Using cmd As New MySqlCommand(
                    "UPDATE tblbarang SET Satuan=@baru WHERE Satuan=@lama", koneksi, transaksi)
                    cmd.Parameters.Add("@baru", MySqlDbType.VarChar, 50).Value = namaBaru
                    cmd.Parameters.Add("@lama", MySqlDbType.VarChar, 50).Value = _satuanLama
                    cmd.ExecuteNonQuery()
                End Using
                transaksi.Commit()
            Catch
                transaksi.Rollback()
                Throw
            End Try
        End Using
    End Using
    MuatGridSatuan()
    MuatSatuan(namaBaru)
    MuatBarang()
    BersihkanBarang()
    BersihkanSatuan()
    MessageBox.Show("Satuan dan referensi barang berhasil diubah.", "Informasi")
  Catch ex As Exception
    TampilkanError("Gagal mengubah satuan", ex)
  End Try
End Sub

Private Sub btnHapusSatuan_Click(sender As Object, e As EventArgs) Handles btnHapusSatuan.Click
  If _noSatuanTerpilih < 0 Then
    MessageBox.Show("Pilih satuan di tabel terlebih dahulu.", "Informasi")
    Return
  End If
  If MessageBox.Show("Hapus satuan '" & _satuanLama & "'?", "Konfirmasi",
            MessageBoxButtons.YesNo, MessageBoxIcon.Question) <> DialogResult.Yes Then Return
  Try
    Using koneksi As MySqlConnection = BukaKoneksi()
      Using cmdCek As New MySqlCommand(
        "SELECT COUNT(*) FROM tblbarang WHERE Satuan=@satuan", koneksi)
        cmdCek.Parameters.Add("@satuan", MySqlDbType.VarChar, 50).Value = _satuanLama
        If Convert.ToInt32(cmdCek.ExecuteScalar()) > 0 Then
          MessageBox.Show("Satuan masih digunakan pada data barang dan tidak dapat dihapus.", "Validasi")
          Return
        End If
      End Using
      Dim sql As String = "DELETE FROM `" & TABLE_SATUAN & "` WHERE `no`=@no"
      Using cmd As New MySqlCommand(sql, koneksi)
        cmd.Parameters.Add("@no", MySqlDbType.Int32).Value = _noSatuanTerpilih
        cmd.ExecuteNonQuery()
      End Using
    End Using
    MuatGridSatuan()
    MuatSatuan()
    BersihkanBarang()
    BersihkanSatuan()
    MessageBox.Show("Satuan berhasil dihapus.", "Informasi")
  Catch ex As Exception
    TampilkanError("Gagal menghapus satuan", ex)
  End Try
End Sub

Private Sub btnBersihSatuan_Click(sender As Object, e As EventArgs) Handles btnBersihSatuan.Click
  Try
    BersihkanSatuan()
  Catch ex As Exception
    TampilkanError("Gagal mengosongkan satuan", ex)
  End Try
End Sub
```

```
    Private Sub dgvSatuan_CellClick(sender As Object, e As DataGridViewCellEventArgs) Handles dgvSatuan.CellClick
        If e.RowIndex < 0 Then Return
        Dim baris As DataGridViewRow = dgvSatuan.Rows(e.RowIndex)
        If baris.IsNewRow Then Return
        _noSatuanTerpilih = Convert.ToInt32(baris.Cells("no").Value)
        _satuanLama = Convert.ToString(baris.Cells("satuan").Value)
        txtNoSatuan.Text =  _noSatuanTerpilih.ToString()
        txtSatuan.Text = _satuanLama
        btnSimpanSatuan.Enabled = False
        btnUbahSatuan.Enabled = True
        btnHapusSatuan.Enabled = True
    End Sub

    Private Sub TampilkanError(kegiatan As String, ex As Exception)
        MessageBox.Show(kegiatan & ": " & ex.Message, "Kesalahan",
                        MessageBoxButtons.OK, MessageBoxIcon.Error)
    End Sub
End Class
```

4. Submission

- A max 5-minute demo video showing at CRUD operation.

Group Assignment – Database & Master Form

Assignment Requirements

1. Database Design & Implementation

- Refine the Entity-Relationship Diagram (ERD) from the previous assignment if needed.
- Create a database based on the group's case study.
- Define tables, attributes, data types, primary keys, and foreign keys.
- Establish appropriate relationships between tables.
- Implement the database using MySQL.

2. Database Connection

- Connect the VB.NET Windows Forms application to the MySQL database.
- Create a reusable database connection module/class.
- Implement and demonstrate a successful database connection.
- Handle basic database connection errors.

3. Master Data Form Design

- Identify and develop at least **two** master data forms relevant to the case study.
- Design forms using appropriate Windows Forms controls.
- Implement basic CRUD operations: Create, Read, Update, and Delete.
- Display existing master data using DataGridView.
- Implement automatic numbering or unique ID generation for master records, where applicable.
- Apply input validation and basic error handling.
- Use ComboBox to retrieve related master data from the database, where applicable.

4. Testing & Demonstration

- Test database connectivity and CRUD operations.

- Verify that data entered through the forms is correctly stored in the database.
- Test input validation, duplicate records, and error handling.
- Demonstrate both master data forms during the class session.

Expected Deliverables

No.	Deliverable	Description
1	Updated ERD	Finalized database structure, including table relationships, PKs, and FKs.
2	Database	MySQL database containing the required tables and sample data
3	Database Connection	Working VB.NET connection module/class.
4	Master Data Forms	At least two functional master data forms with CRUD operations.
5	Source Code	Updated VB.NET Windows Forms project.
6	Documentation	Submit a recorded video demonstrating the completed master data forms, including database connectivity, CRUD operations, input validation, and error handling. Every group member must appear on camera and participate in explaining or demonstrating their contribution to the project.

Minimum Completion Target: Each group must have a working MySQL database, a successful database connection, and at least two functional master data forms connected to the database.

Submission

- Upload video link to ILMU2